Customer
Payment Page (app/booking/payment/page.tsx)
VNPay Return Page (app/booking/payment/return/page.tsx)
Booking Detail Page (app/booking/[id]/page.tsx)
Router (next/navigation)
Booking API (lib/api/booking.ts)
Payment Controller (booking-service)
Payment Service (booking-service)
Booking Service (booking-service)
Hotel Availability Client (booking-service)
Availability Controller (hotel-service)
Availability Service (hotel-service)
VnPay Util (booking-service)
Loyalty Points Client (booking-service)
Hotel Event Publisher (booking-service)
BookingDB
HotelDB
RabbitMQ
VNPay

25. Select VNPay payment method
26. Confirm payment
27. validate bookingId and timeRemaining

alt [payment validation failed]
28. Display payment validation error message
[payment validation passed]
29. check paymentMethod
30. bookingApi.initVnPay(bookingId)
30a. POST /api/payments/vnpay/create?bookingId=...
30b. initVnPayPayment(bookingId, clientIp)
30c. bookingRepository.findById(bookingId)
30d. Booking
30e. paymentRepository.findFirstByBookingIdOrderByCreatedAtDésc(bookingId)
30f. latest payment
30g. ensureActiveHold(bookingId)
30h. getHold(holdId)
30i. GET /api/v1/availability/hold/{holdId}
30j. getHold(holdId)
30k. roomHoldRepository.findById(holdId)
30l. RoomHold
30m. HoldResponse
30n. HoldResponse
30o. HoldResponse
30p. Booking
30q. bookingRepository.save(PENDING_PAYMENT)
30r. paymentRepository.save(payment PENDING)
30s. buildQuery(params)
30t. query
30u. hmacSHA512(hashSecret, query)
30v. secureHash
30w. paymentRepository.save(payment with vnpayOrderId)
30x. saved payment
30y. VnPayInitResponse
30z. VnPayInitResponse
30aa. redirectUrl
31. window.location.href = redirectUrl
32. process online payment
33. redirect(returnUrl)
34. bookingApi.processVnPayReturn(qs)
34a. GET /api/payments/vnpay/return?...
34b. buildQuery(paramsForHash)
34c. data
34d. hmacSHA512(hashSecret, data)
34e. expected
34f. processVNPayCallback(orderId, responseCode, transactionNo)
34g. paymentRepository.findByVnpayOrderId(orderId)
34h. Payment

alt [responseCode != "00"]
35. save payment FAILED and booking PENDING
35a. failed payment and pending booking saved
35b. PaymentResponse
35c. PaymentResponse
35d. PaymentResponse
36. Display payment failure message

alt [retry payment]
37. Retry payment
37a. push("/booking/payment?bookingId=...")
38. Display payment page with existing booking
38a. Confirm retry payment
38b. bookingApi.initVnPay(bookingId)
38c. POST /api/payments/vnpay/create?bookingId=...
38d. initVnPayPayment(bookingId, clientIp)
38e. ensureActiveHold(bookingId)
38f. getHold(holdId)
38g. GET /api/v1/availability/hold/{holdId}
38h. getHold(holdId)
38i. HoldResponse
38j. HoldResponse
38k. HoldResponse
39. Is room hold still valid?

alt [Hold expired or not HELD]
40. bookingRepository.save(CANCELLED)
40a. saved booking
40b. publishBookingStatus(event)
40c. convertAndSend("booking.status.changed", event)
40d. return
40e. throw IllegalStateException
40f. error
40g. 400 Bad Request
40h. throw Error
41. Display hold expired message
[Hold still valid]
38l. Booking
42. paymentRepository.save(new VNPAY payment)
42b. saved payment
42c. buildQuery(params)
42d. hmacSHA512(hashSecret, query)
42e. VnPayInitResponse
42f. VnPayInitResponse
42g. redirectUrl
43. window.location.href = redirectUrl
44. process online payment
45. redirect(returnUrl)
46. bookingApi.processVnPayReturn(qs)
46a. GET /api/payments/vnpay/return?...
46b. processVNPayCallback(orderId, responseCode, transactionNo)
46c. paymentRepository.findByVnpayOrderId(orderId)
46d. Payment

alt [retry responseCode != "00"]
47. save payment FAILED and booking PENDING
47a. failed payment and pending booking saved
47b. PaymentResponse
47c. PaymentResponse
47d. PaymentResponse
48. Display payment failure message
[retry responseCode == "00"]
49. save payment COMPLETED
50. save booking CONFIRMED
50a. completed retry payment and confirmed booking saved
51. confirmBookingHold(bookingId)
51a. confirmHold(holdId)
51b. POST /api/v1/availability/hold/{holdId}/confirm
51c. confirmHold(holdId)
51d. update held/booked and hold status
51e. saved
51f. HoldResponse
51g. HoldResponse
51h. return
51i. return
52. usePoints(request) / earnPoints(userId, bookingId, totalAmount)
52a. publishPaymentCompleted(event)
52b. convertAndSend("payment.completed", event)
52c. publishBookingStatus(event)
52d. convertAndSend("booking.status.changed", event)
52e. return
52g. PaymentResponse
52h. PaymentResponse
52i. PaymentResponse
53. Display VNPay success confirmation
64. View booking confirmation
64a. router.push("/booking/{bookingId}")
64b. Display booking detail page

[no retry]
54. Keep failed result page
55. Display failed payment status

[responseCode == "00"]
56. save payment COMPLETED
57. save booking CONFIRMED
57a. completed payment and confirmed booking saved
58. confirmBookingHold(bookingId)
58a. confirmHold(holdId)
58b. POST /api/v1/availability/hold/{holdId}/confirm
58c. confirmHold(holdId)
58d. update held/booked and hold status
58e. saved
58f. HoldResponse
58g. HoldResponse
58h. return
58i. return
59. usePoints(request) / earnPoints(userId, bookingId, totalAmount)
59a. publishPaymentCompleted(event)
59b. convertAndSend("payment.completed", event)
59c. publishBookingStatus(event)
59d. convertAndSend("booking.status.changed", event)
59e. return
59g. PaymentResponse
59h. PaymentResponse
59i. PaymentResponse
60. Display VNPay success confirmation
64. View booking confirmation
64a. router.push("/booking/{bookingId}")
64b. Display booking detail page

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